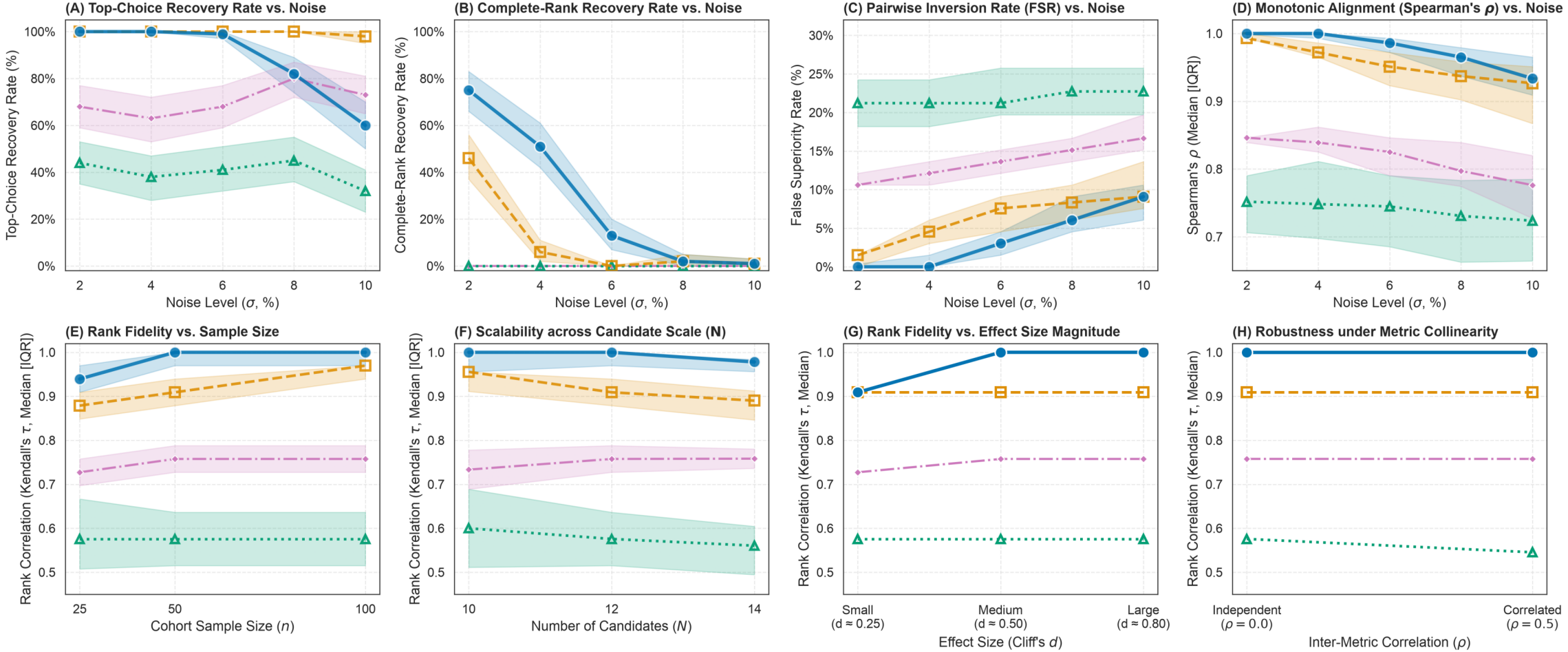


Benchmark Validation Overview

Default parameters (unless otherwise specified): $N = 12$, $n = 50$, Cliff's $d \approx 0.50$, $\rho = 0.0$, $K = 100$ runs per point

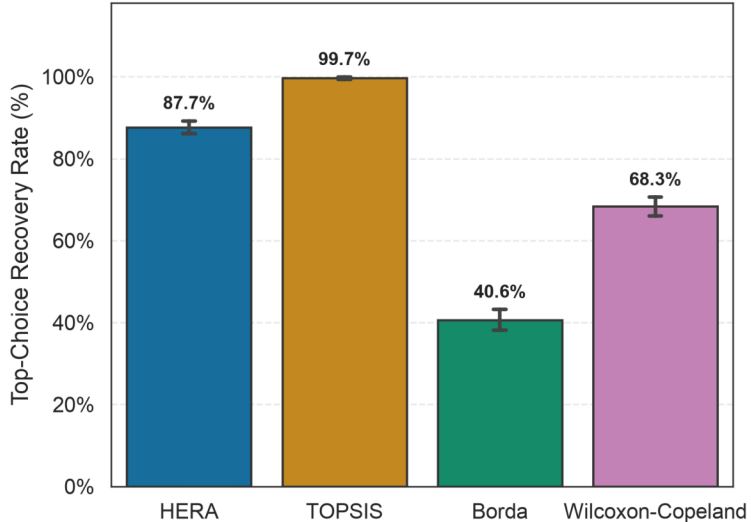


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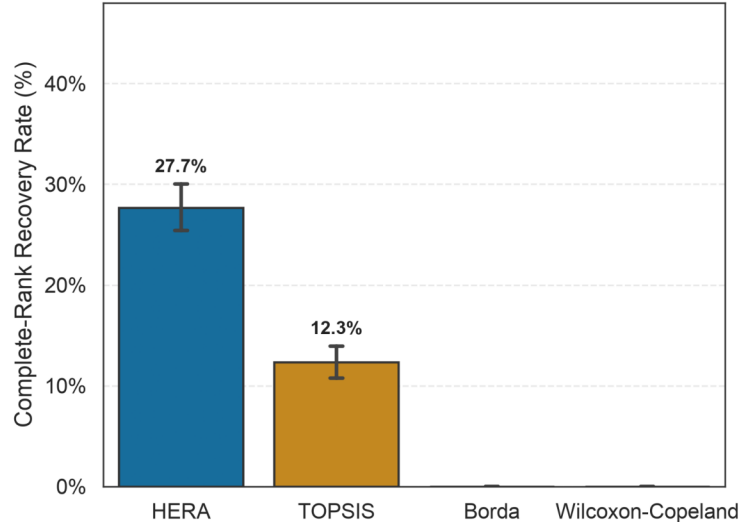
Pooled Distribution across Core Conditions

$N \in \{10, 12, 14\}$, $\sigma \in \{2\%, 4\%, 6\%, 8\%, 10\%\}$, $n = 50$, $\rho = 0.0$; $K = 1499$ runs per method

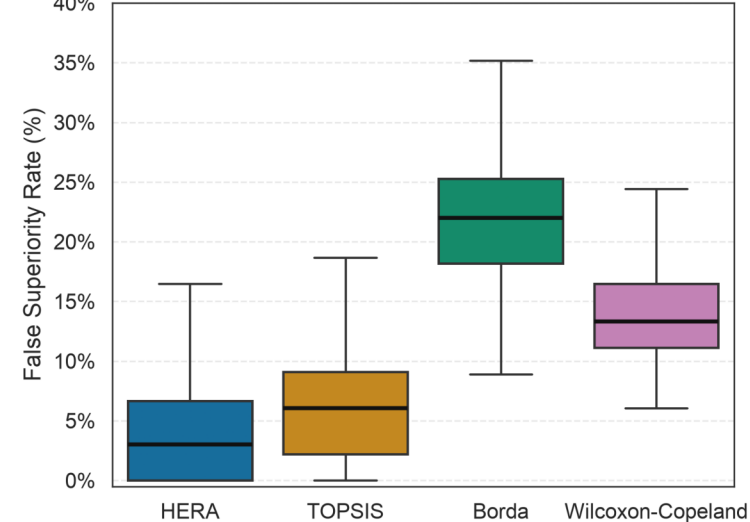
(A) Top-Choice Recovery Rate



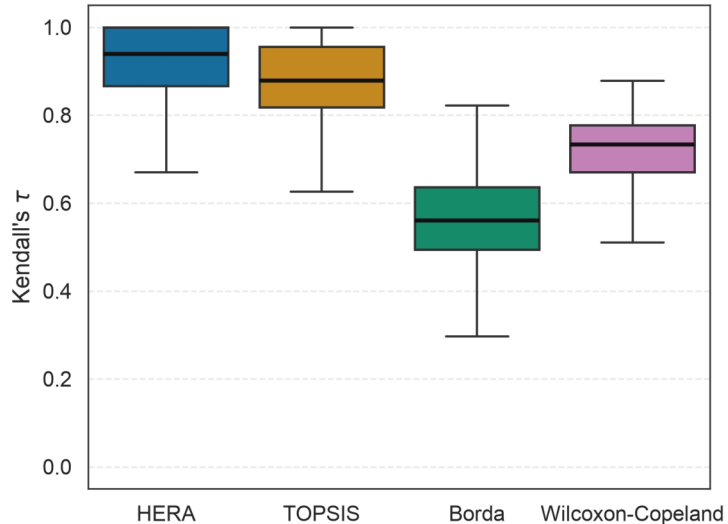
(B) Complete-Rank Recovery Rate



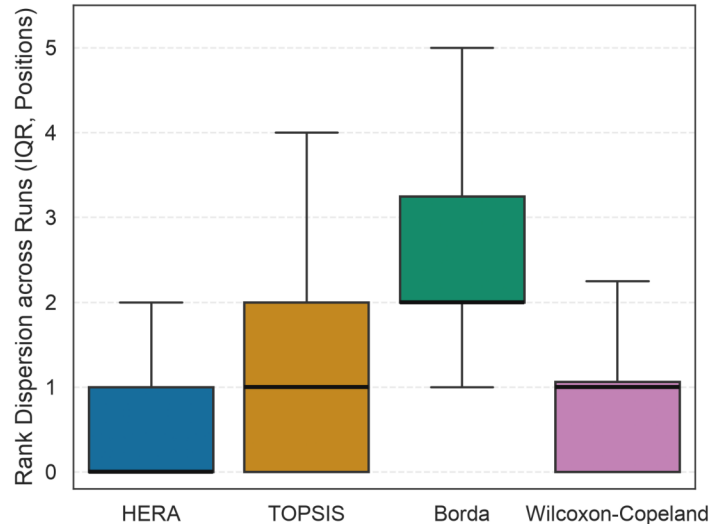
(C) Pairwise Inversion Rate (False Superiority)



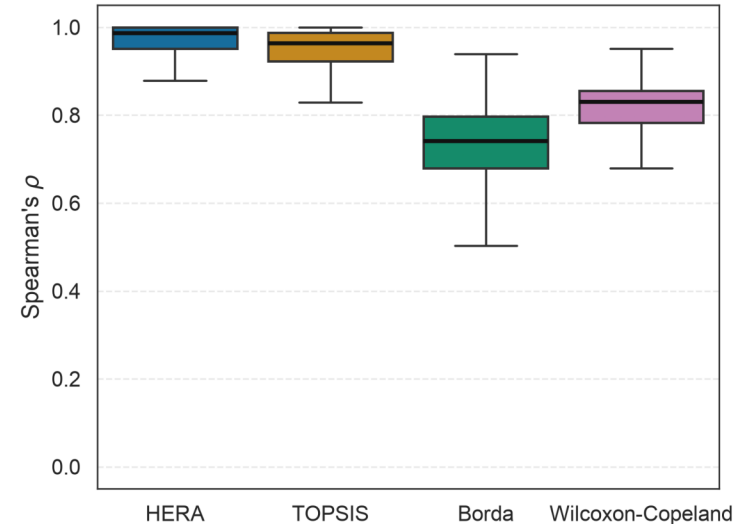
(D) Rank Correlation Fidelity (Kendall's τ)



(E) Empirical Rank Stability (Candidate-IQR)

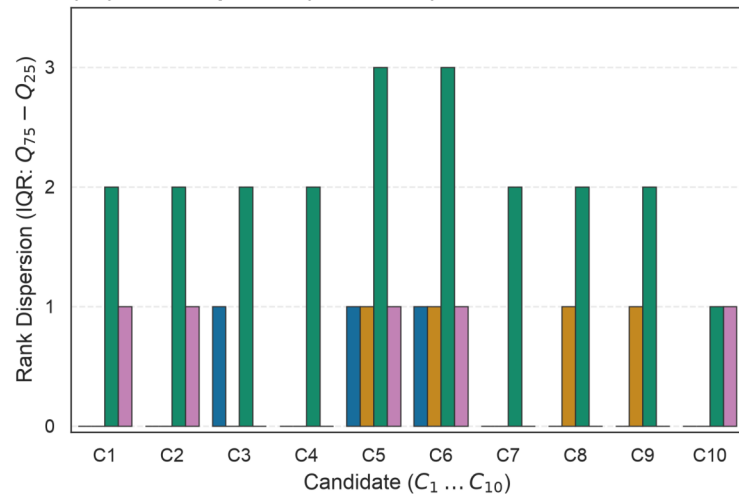


(F) Monotonic Alignment (Spearman's ρ)

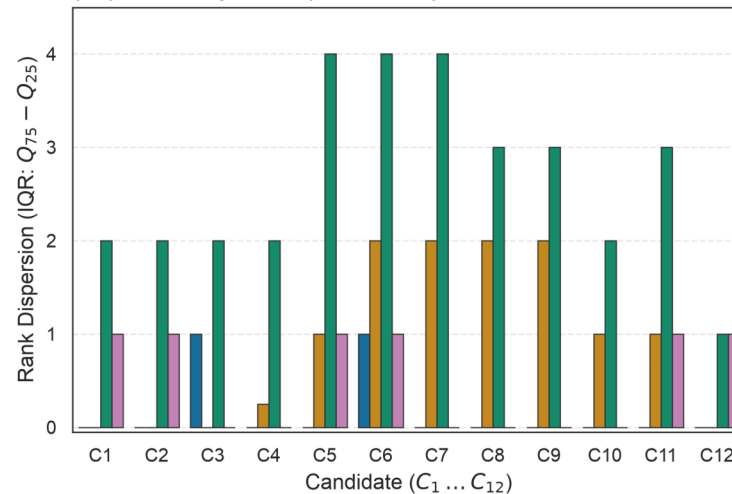


Candidate-Level Rank Stability Across Scales
 $N \in \{10, 12, 14\}$, $n = 50$, $\sigma = 4\%$, $\rho = 0.0$, $K = 100$ runs per point

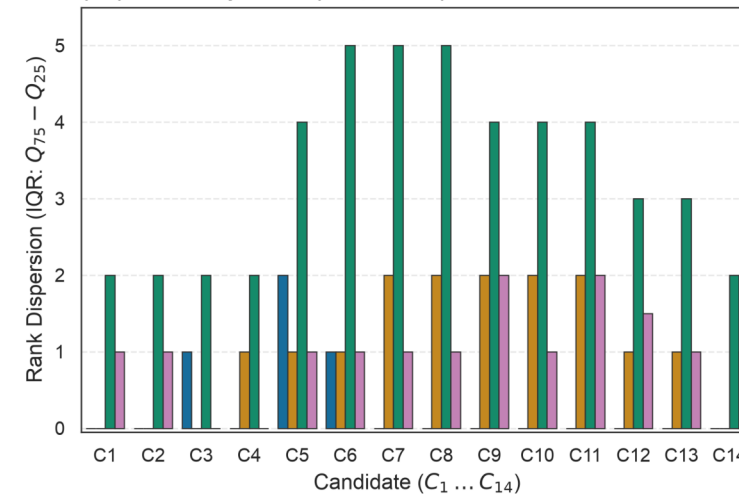
(A1) Rank Dispersion (IQR, $N = 10$)



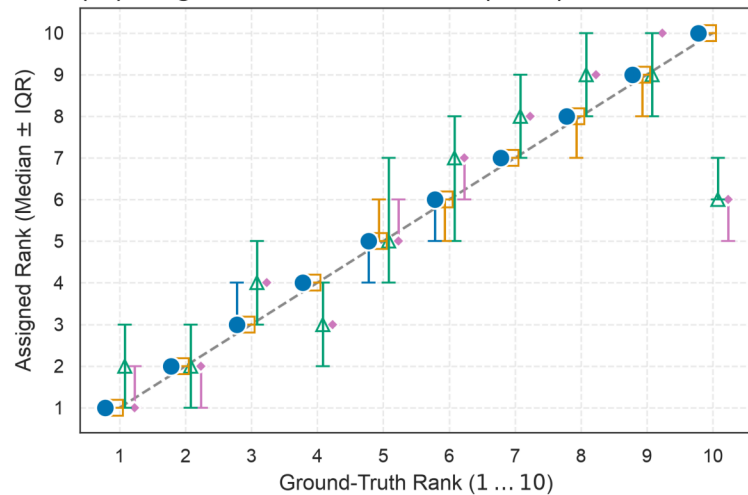
(A2) Rank Dispersion (IQR, $N = 12$)



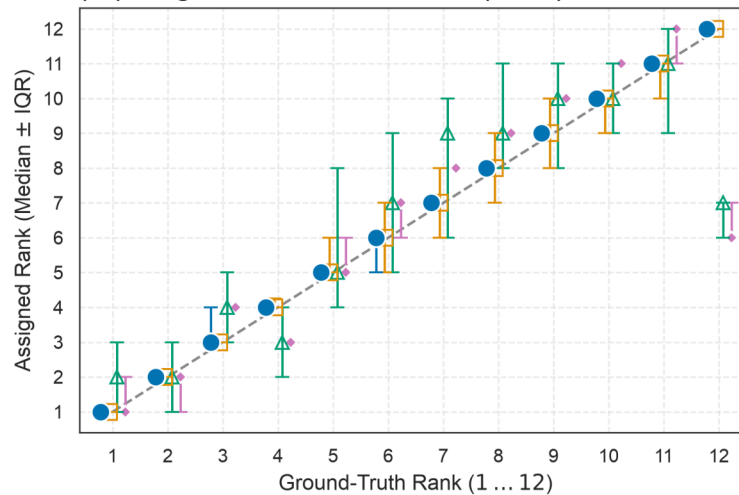
(A3) Rank Dispersion (IQR, $N = 14$)



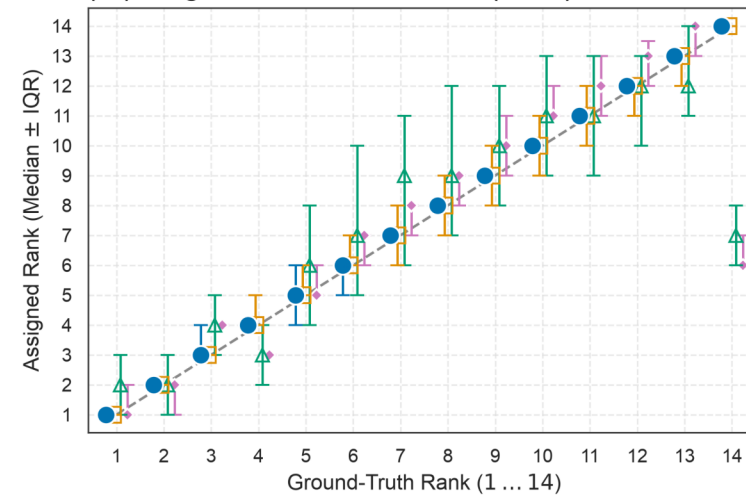
(B1) Assigned Rank vs. Ground Truth ($N = 10$)



(B2) Assigned Rank vs. Ground Truth ($N = 12$)



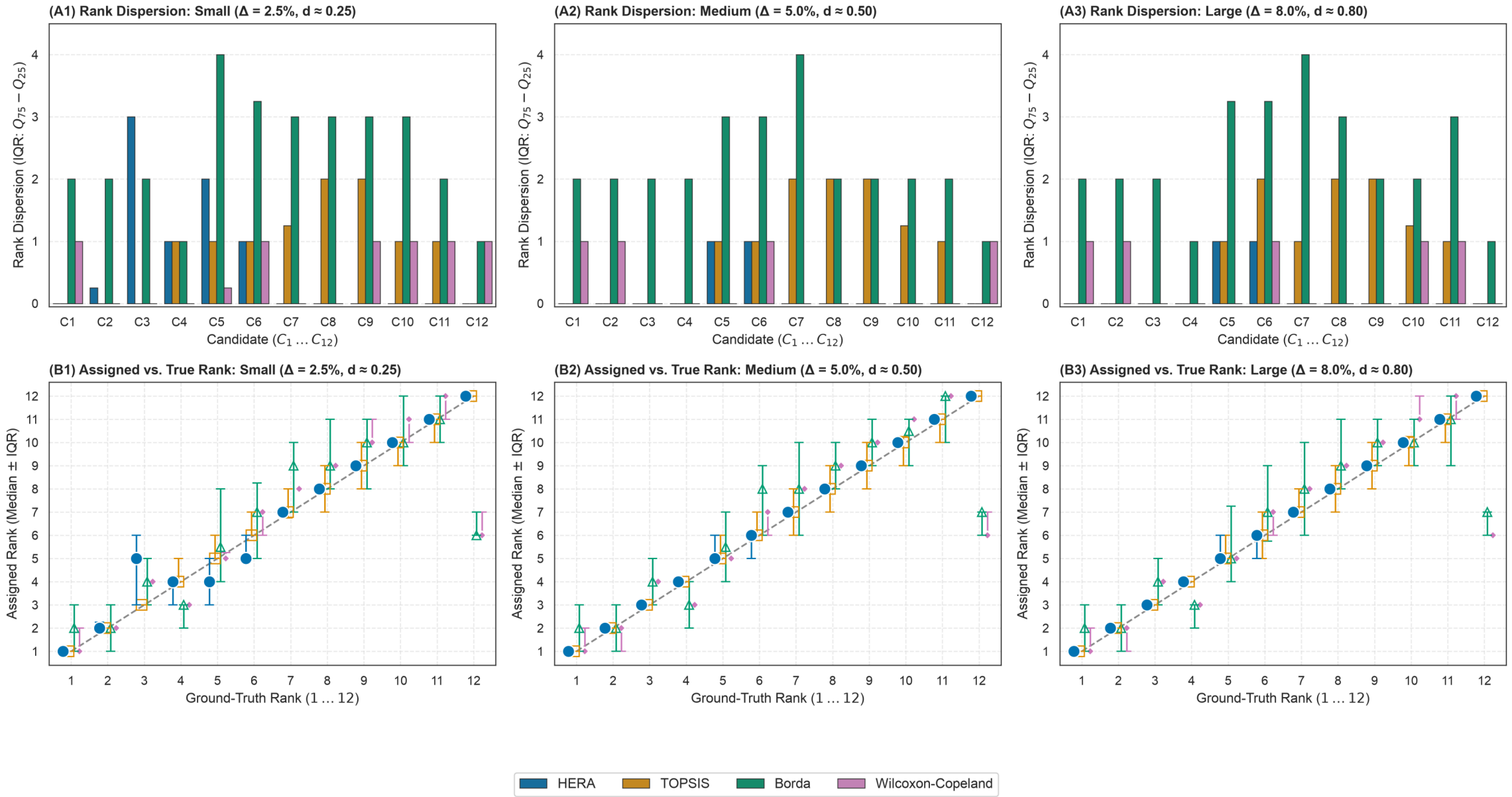
(B3) Assigned Rank vs. Ground Truth ($N = 14$)



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Effect Size Sensitivity: Candidate-Level Rank Dispersion & Fidelity

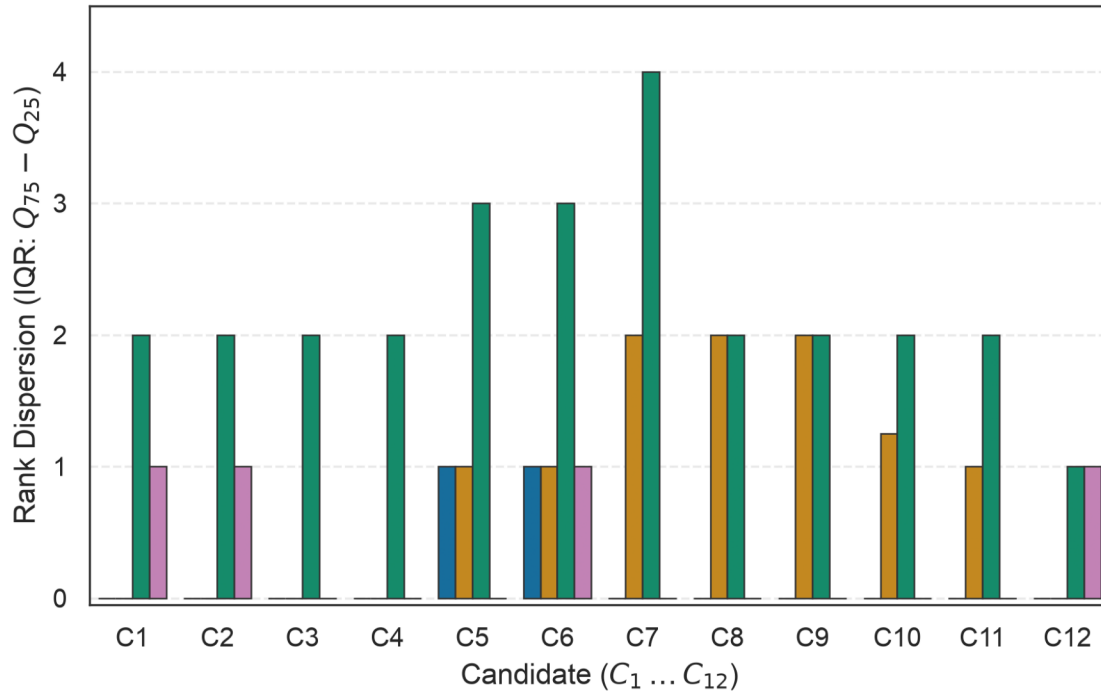
N = 12, n = 50, σ = 4%, p = 0.0, K = 100 runs per condition



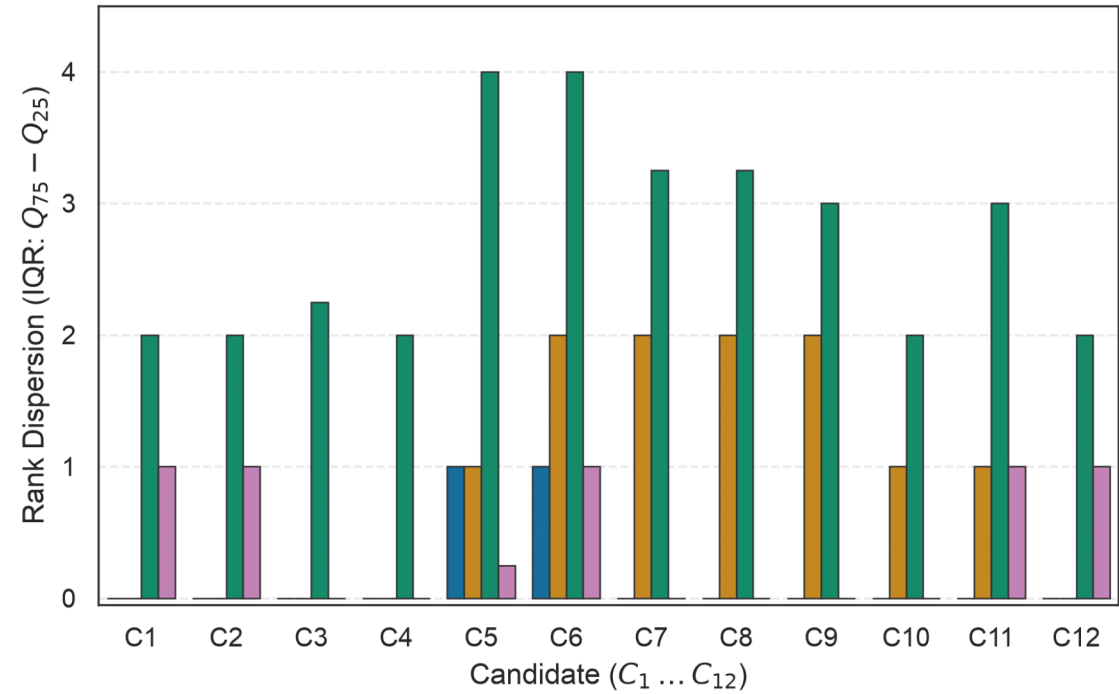
Inter-Metric Collinearity Sensitivity: Rank Dispersion & Fidelity

N = 12, n = 50, $\sigma = 4\%$, Cliff's $d \approx 0.50$, K = 100 runs per condition

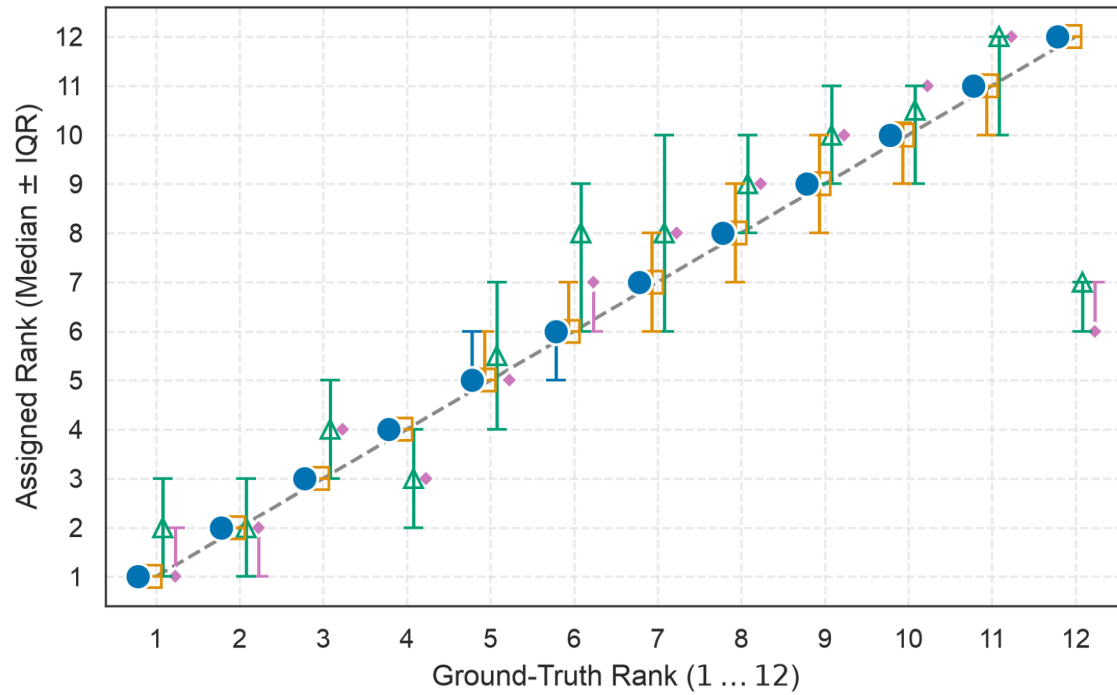
(A1) Rank Dispersion: Independent Criteria ($\rho = 0.0$)



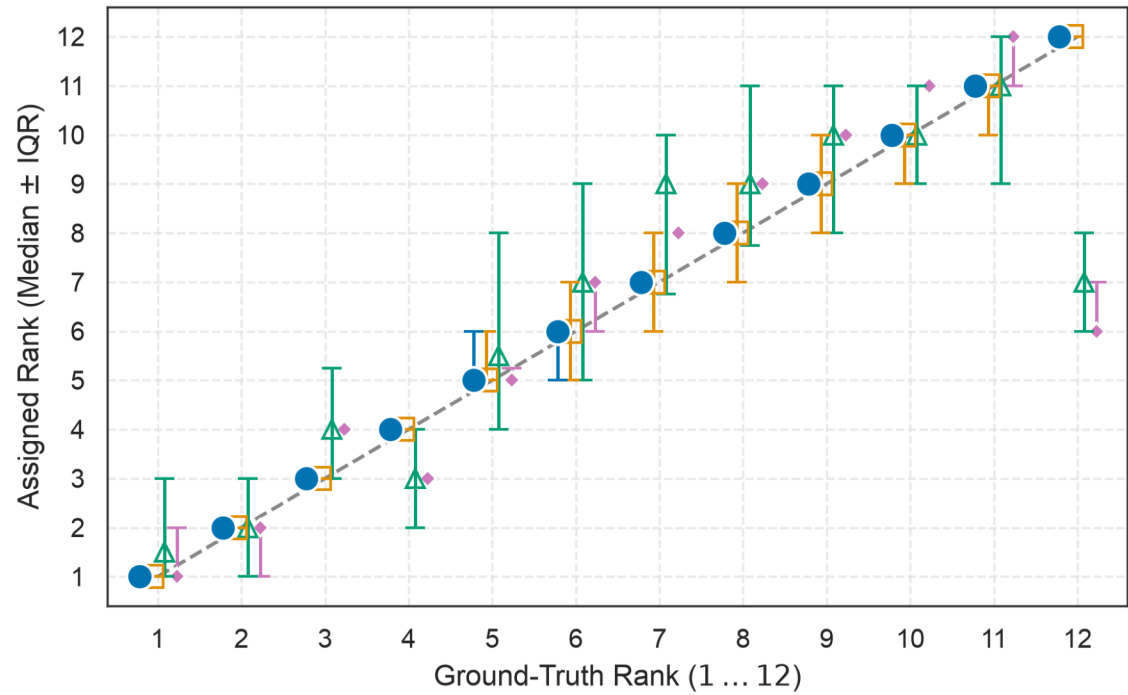
(A2) Rank Dispersion: Correlated Criteria ($\rho = 0.5$)



(B1) Assigned Rank vs. Ground Truth: Independent Criteria ($\rho = 0.0$)



(B2) Assigned Rank vs. Ground Truth: Correlated Criteria ($\rho = 0.5$)



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